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MILITARY SPECIFICATION

COMPRESSORS, RECIPROCATING AND ROTARY

GASOLINE-ENGINE-DRIVEN, AIR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers gasoline engine-driven, garage type, reciprocating and rotary air compressors.

1.2 Classification. The compressors shall be of the following types, classes, and sizes as specified (see 6.2). Class 2 shall be limited to the 5-cubic-feet-per-minute (cfm) size.

Type I - Reciprocating.
Type II - Rotary.
Class 1 - Tank-mounted.
Class 2 - Handtruck mounted.
Class 3 - Caster mounted.
Class 4 - Trailer mounted.
Size - 5 cfm at 175 psig.
Size - 15 cfm at 175 psig.
Size - 25 cfm at 175 psig.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research and Development Center, ATTN: STRBE-DS, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

FEDERAL

FF-C-88	- Casters, Rigid and Swivel, Industrial, Heavy-Duty.
GG-G-91	- Gages, Tire Pressure, Self-Contained and Inflator-Gage, Pneumatic Tire.
ZZ-B-190	- Belts, V: Engine Accessory Drive; Minus 40° F, General Specification for.
ZZ-I-550	- Inner Tube, Pneumatic Tire.
ZZ-T-410	- Tire, Pneumatic, Industrial.

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MIL-V-173	- Varnish, Moisture- and Fungus-Resistant (for Treatment of Communications, Electronic, and Associated Equipment).
MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-G-3056	- Gasoline, Automotive, Combat.
MIL-F-3541	- Fittings, Lubrication.
MIL-C-3600	- Compressors, Rotary, Power-Driven; and Compressors, Reciprocating, Power-Driven: Air and Gas (Except Oxygen and Refrigerant), Packaging of.
MIL-G-3859	- Grease, Guns, Hand-Operated, Lever, Push and Screw Type.
MIL-H-4607	- Heater, Engine and Shelter, Ground, Type H-1.
MIL-H-7365	- Hose, Air Duct, for Ground Heaters.
MIL-G-10924	- Grease, Automotive and Artillery.
MIL-B-11040	- Belt, V: Engine Accessory Drive.
MIL-C-13489	- Cover and End Curtains; Cargo Body (for Military Vehicles).
MIL-R-14328	- Rubber Sheet: Synthetic, Medium Soft, General-Purpose Gasket Material (for Extreme Climatic Conditions).
MIL-G-20241	- Gasket, Material, Wool Felt, Impregnated, Adhesive, Pressure-Sensitive.
MIL-G-23827	- Grease, Aircraft and Instrument, Gear and Actuator Screw.
MIL-L-46167	- Lubricating Oil, Internal Combustion Engine, Arctic.
MIL-C-46168	- Coating, Aliphatic Polyurethane, Chemical Agent Resistant.
MIL-C-52437	- Chassis Trailer Single and Multi-Axle.
MIL-S-81733	- Sealing and Coating Compound, Corrosion Inhibitive.

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STANDARDS

FEDERAL

FED-STD-H28	- Screw-Thread Standards For Federal Services.

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MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-209	- Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.
MIL-STD-461	- Electromagnetic Interference Characteristics Requirements for Equipment.
MIL-STD-462	- Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-838	- Lubrication of Military Equipment.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1226	- Engine, Gasoline: Air-Cooled, 1-1/2 BHP, 4-Cycle, Military Design, Model 1A08, Installation Procedures.
MIL-STD-1300	- Engine, Gasoline, Air-Cooled, 6 BHP, Military Design, Model 4A032, Installation Procedures.
MIL-STD-1410	- Methods for Selection of Industrial Engines for End Item Application.
MIL-STD-1474	- Noise Limits For Army Materiel.
MS35644	- Filler Neck Assembly, Fuel Tank-Military Vehicles.

HANDBOOK

MILITARY

MIL-HDBK-113	- Guide for the Selection of Lubricants, Functional Fluids, Preservatives, Preservatives and Specialty Products for Use in Ground Equipment Systems.

2.1.2 Other Government document, drawing, and publication. The following other Government document, drawing, and publication form a part of this specification to the extent specified herein.

ENVIRONMENTAL PROTECTION AGENCY (EPA)

40 CFR 204 - Noise Emission Standards for Construction Equipment.

(Application for copies should be addressed to the Superintendent of Documents, US Government Printing Office, Washington, DC 20402).

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DRAWING

ME

13208E5819 - Kit Assembly, Starter.

(Copies of specifications, standards, and drawings required by contractors in connection with specfic procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY OF MECHNICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Section VIII.
Performance Test Code 9 - Displacement Compressors, Vacuum Pumps and Blowers.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE J492 - Rivets and Riveting.
SAE J833 - USA Human Physical Dimensions, Recommended Practices.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

SOUTH COAST AIR QUALITY MANAGEMENT DISTRICT

Rule 442 - Usage of Solvents.

(Application for copies should be addressed to the South Coast Air Quality Management District, 9150 Flair Drive, El Monte, CA 91731).

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The air compressor shall be of the commercial classification "garage type" and shall consist of either an air-cooled piston compressor unit or air-oil cooled, rotary compressor unit with an air reservoir, air hose and power unit.

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3.2 First article. The contractor shall furnish one or more compressors for examination and test within the time frame specified (see 6.2), to prove prior to starting production that his production methods and choice of design detail will produce compressors that comply with the requirements of this specification. Examination and tests shall be as specified in section 4; and unless otherwise specified, all examination and tests shall be conducted by the contractor, subject to surveillance and approval by the Government (see 6.3). When specified (see 6.2), the Government may conduct any or all of the first article examination and tests, as specified (see 6.2).

3.3 Winterized first article. When winterized compressors are specified (see 3.25) in addition to nonwinterized compressors, two first article models shall be furnished for examination and test. One of these first article models shall be a nonwinterized compressor and one shall be a winterized compressor.

3.4 Material. Material shall be as specified herein and on the drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.12).

3.4.1 Material deterioration and control. The compressor shall be fabricated from compatible materials, inherently corrosion and deterioration resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable storage and operating environments to which the compressor may be exposed.

3.4.2 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the air compressor may be newly fabricated from recovered materials provided the air compressor produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be used.

3.5 Dissimilar metals. Dissimilar metals, as defined in MIL-STD-889, shall be electrically insulated from one another to minimize or prevent galvanic corrosion. Insulation may be provided by an insulating barrier such as a corrosion inhibiting sealant conforming to MIL-S-81733 or chromate tape conforming to MIL-G-20241. Protection against corrosion may also be obtained by exclusion of the electrolyte if feasible.

3.5.1 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with components and sub-components, and shall make information available, upon request, to the contracting officer or designated representative.

3.6 Operating temperatures. The compressors shall perform as specified herein in any ambient temperature from 120° F. to minus 25° F. for nonwinterized compressors; and from 120° F. to minus 50° F. for winterized compressors (see 3.25).

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3.7 Safety and human factors.

3.7.1 Safety. All rotating or reciprocating parts and other parts subject to high operational temperatures that are of such a nature or are so located as to be a hazard to operating personnel, shall be guarded or insulated to the extent necessary to eliminate the hazard. Protective devices shall not impair the operating functions.

3.7.2 Human factors. The characteristics of the compressor shall provide for operation by personnel ranging from the small man (5th percentile), through the large man (95th percentile), in accordance with SAE J833, including arctic clothed personnel when winterization is specified.

3.7.3 Noise limits.

3.7.3.1 Sound limits. The noise level when delivering rated capacity shall not exceed 85 dB(A) at 15 feet when tested in accordance with 4.5.2.11. The distance from the control panel at which the noise level is 85 dB(A) shall be determined. That distance shall be shown on warning plates mounted on both sides of the compressor, stating "Warning. Ear Protection Required within _____ feet". The warning shall be legible from the distance where the level is met.

3.7.3.2 Personnel noise. If noise level exceeds 85 dB(A) at the control station when tested in accordance with 4.5.2.11, a feasibility study in accordance with MIL-STD-1474, 5.1.1.2, and cost estimate for reducing noise level to 85 dB(A) shall be provided.

3.7.3.3 Exterior noise. The compressor shall meet the requirements of EPA 40 CFR 204, when tested in accordance with 4.5.2.11.1

3.8 Endurance and maintainability.

3.8.1 Endurance. The compressor shall produce rated capacity at rated pressure for 500 hours in operating periods of not less than 8 hours. Operating periods of not less than 4 hours during high and low temperature tests may be counted (see 4.5.2.6, 4.5.2.7, 4.5.2.18 and 4.5.2.19).

3.8.2 Maintainability. The compressor shall operate as specified herein without repair or maintenance other than the contractor's recommended normal schedule adjustments and servicing, as established by a maintenance schedule prepared and submitted by the contractor prior to test. All major assemblies and installed attachments shall be accessible for maintenance repair and replacement without removal of other major assemblies and installed attachments not normally removed. Cover plates which must be removed for component adjustment shall be equipped with quick-disconnect fastenings. All fasteners shall be of corrosion-resistant material or shall be treated to be corrosion resistant. All screw threads shall be in accordance with FED-STD-H28. Maximum use shall be made of interchangeable hardware and fastening devices. Means for drainage of lubricants, fuel, and condensate traps shall be in an accessible location and

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shall drain without splashing on any component or on operating personnel. The drain outlets shall be so located to facilitate complete drainage. Each maintenance, assembly or disassembly operation performed as a result of testing shall be accomplished by not more than two men using common tools and special tools furnished with the compressor.

3.9 Air hose. The air hose shall be the commercial heavy duty type 5/16 inch ID, with minimum rated working pressure of not less than 200 psi. Unless otherwise specified (see 6.2), the hose shall be not less than 50 feet long. Air hose couplings shall be internal swivel fitted, having a 1/4-inch, standard, straight-pipe thread and a 30-degree tapered seat and shall be fitted at one end with an inflator gage conforming to GG-G-91, type III, style 3.

3.10 Intake air filter and silencer. The compressors shall be provided with a dry-type combination intake air filter and silencer and shall have a rain-shield or a similar protective device.

3.11 Electromagnetic interference. Unless otherwise specified (see 6.2), the electromagnetic interference requirments shall be in accordance with MIL-STD-461, class III C.

3.12 Lubrication. All surfaces requiring lubrication shall be provided with a means for lubricating.

3.12.1 Lubricants. The compressors shall operate as specified herein when lubricated with military lubricants (see 6.4).

3.12.2 Lubrication fittings. Lubrication fittings shall conform to MIL-F-3541, type I, II or III. Fittings shall be located in protected positions. Fittings shall be accessible to a grease gun conforming to MIL-G-3859, type II. Accessibility to fittings shall be provided without the removal or adjustment of accessories or parts. Panels and plates equipped with hand-operable, quick-disconnect fasteners may be removed to provide accessibility to lubrication fittings.

3.12.3 Pressure-release device. A pressure-release device shall be provided where the use of pressure lubricating equipment will damage grease seals or other parts.

3.12.4 Enclosures with integral reservoir of lubricant. Enclosures, such as gear cases and transmission housings which contain a reservoir of lubricants for lubrication of the parts enclosed, shall be equipped with dipsticks, check plugs not less than 1/2-inch pipe size, or sight gages to determine the level of the lubricant. Each enclosure shall be equipped with a drain plug. Each enclosure shall be equipped with a means for filling the enclosure with lubricant. The drain plug shall be located so that removal of the drain plug will result in complete drainage of the lubricant from the enclosure. Drainage shall be accessible for draining to a container when the compressor is in its normal position. Integral tubes or troughs may be used to convey lubrication from the drain outlet. Accessibility to the drain plug, the filling means, and the

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lubricant-level checking device shall be obtained without the removal or adjustment of accessories or parts. Engine-housing side panels and plates equipped with hand-operable, quick-disconnect fasteners may be removed to facilitate this accessibility.

3.12.5 Delivery lubrication service. When delivered, all components, except those for which preservatives are specified in 5.1, shall be lubricated. Grease lubrication, excluding lubrication of sealed bearings, shall be with grease conforming to MIL-G-10924 or MIL-G-23827. Unless otherwise specified herein, all components shall be filled to the operating level with military oils designated for use in the temperature range as specified herein. Each lubricated component shall be tagged in a conspicuous place to indicate the temperature range and grades of oils and greases used.

3.13 Fungus and moisture resistance. The electrical circuitry, including all components and connections except as specified below, shall be protected from the effects of moisture and fungus growth by an overall treatment with a varnish conforming to MIL-V-173, composition I or II as applicable, 7 percent salicylanilide (by weight) based on the nonvolatile content of the varnish:

a. Components or circuit elements that are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.
b. Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a minimum dry-film thickness of 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

3.13.1 Composition. Composition II shall be used only in the case when local air pollution regulations in the application of varnish precludes the use of compositon I. When composition II is used, the contractor shall provide evidence to the Government that the use of composition II is required, and shall certify that composition II material complies with Rule 442 - Usage of Solvents.

3.14 Identification marking. The compressor shall be identified in accordance with MIL-STD-130. The marking shall be applied to the compressor on identificaton plates conforming to MIL-P-514, type I, style 1, composition C, of type I, grade A, class 1 material. Plates shall be attached by screws, bolts or rivets in a conspicuous, protected location.

3.15 Instruction plates. Each compressor shall be equipped with instruction plates or diagrams, including warnings and cautions, describing any special or important procedures to be followed in assembling, operating or servicing the compressor. Instruction plates shall conform to MIL-P-514, type III, compositon C, of type I, grade A, class 1 material. Plates shall be attached by screws, bolts or rivets in a conspicuous protected location.

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3.16 Treatment and painting. The compressor, components, and parts painted in commercial practice shall be cleaned, treated and painted in accordance with MIL-T-704, type F or G as applicable. Unless otherwise specified (see 6.2), top coat color shall be camouflage green 383 conforming to MIL-C-46168.

3.17 Government-furnished property. Except as specified for size 25 CFM compressors and unless otherwise specified (see 6.2), the following property in quantitites indicated will be furnished by the Government for the applicable size compressor (see 6.5):

Item No.	Description	Identification	Compressor Size	Quantity for each compressor
1	Military design 1-1/2-hp engine	Model 1A08 NSN 2805-00-068-7510	5-cfm	1
2	Military design 6-hp engine	Model 4A032 NSN 2805-00-068-7512	15-cfm	1
3	Generator 6-hp engine	NSN 2920-00-882-2461	15-cfm	1
4	Starter 6-hp engine	NSN 2920-00-882-3401	15-cfm	1

3.18 Government-loaned property. Unless otherwise specified (see 6.2), the following property in the quantity indicated will be loaned by the Government (see 6.6).

Description	Identification	Quantity
Heater, engine and shelter, ground Type H-1	NSN 4520-00-203-0224	1 ea

3.19 Type I, reciprocating.

3.19.1 Compressor. The type I compressor shall deliver the capacities of free air (see 6.7) specified in table I at a pressure of 175 pound per square inch gage, (psig) as determined by ASME Performance Test Code 9 - Displacement Compressors, Vacuum Pumps, and Blowers. The components of the compressor shall be balanced so that when the compressor is operated in conjunction with the gasoline engine, vibration shall be minimized. The compressor shall not exceed the contractor's recommended continuous operating speed. The compressor shall be air cooled. The cylinder heads or cylinder assembly (if cast integral) shall be removable. The intake and exhaust valves shall be removable. A fan shall be provided for cooling.

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TABLE I. Compressor characteristics.

Size	Minimum capacity (free air)	Maximum operating pressure	Rated discharge pressure
(cfm)	(cfm)	(psig)	(psig)
5	5	200	175
15	15	200	175
25	25	200	175

3.19.2 Intercooler and aftercooler. When a multiple-stage compressor is furnished, an intercooler shall cool the air between the pressure stages and shall be located so as to be cooled by the air blast from the fan. A spring-loaded-type pressure-relief valve shall be installed between the pressure stages to protect the low-pressure cylinder in case of failure of the high-pressure cylinder valves. The temperature measured at the discharge of the air receiver shall not exceed ambient plus 100° F. at a pressure of 200 psig. If necessary to meet the temperature requirements specified herein, an after-cooler may be provided between the last stage of the compressor and the air receiver tank. Aftercoolers shall be air cooled.

3.19.3 Pressure tubing and fittings. All pressure tubing and fittings shall be rated to withstand 1-1/2 times their normal working pressure.

3.19.4 Drive. The drive shall consist of matched multiple V-belts and grooved pulleys on the power unit and compressor unit. Except for the winterized compressor (see 3.25.7), all drive belts shall conform to ZZ-B-190. A means for belt removal and adjustment shall be provided.

3.20 Type II, rotary.

3.20.1 Compressor. The type II compressor shall deliver the capacities of free air specified in table I at a pressure of 175 psig as determined by the ASME Performance Test Code 9 - Displacement Compressors, Vacuum Pumps, and Blowers. The components of the compressor shall be balanced so that when the compressor is operating in conjunction with the gasoline engine, vibration shall be minimized. The compressor shall not exceed the contractor's recommended continuous operating speed. The compressor unit shall be positive displacement and air-oil cooled. A means shall be provided to remove and recover the oil from the discharge air. When a filter-type separator is used, it shall be accessible for examination, cleaning and replacment without removal of other parts or components. To prevent hydraulic lock upon starting, a means shall be provided in the stator assembly to allow accumulated oil to pass into the sump. A pressure gage shall be provided between the compressor unit and the separator discharge to indicate the pressure in the compressor. A nonreturn valve shall be provided either in the oil separator or in the discharge from the separator to prevent reverse flow of air from the receiver.

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3.20.2 Pressure tubing and fittings. All pressure tubing and fittings shall be capable of withstanding 1-1/2 times their normal working pressure.

3.20.3 Drive. The type II compressor shall be either direct drive or multiple V-belt drive. The multiple V-belt drive shall conform to 3.19.4. When a direct drive is used, a flexible coupling shall be installed between the power unit and compressor unit. The mounting base for the power unit shall have a means for belt removal and adjustment when V-belt drive is used.

3.21 Class 1, tank-mounted. The compressor unit and power unit shall be mounted on a base attached to the air-receiver tank.

3.21.1 Air-receiver tank. The air-receiver tank for the class 1 compressor shall be of the horizontal type and shall be constructed in accordance with the ASME Boiler and Pressure Vessel Code, Section VIII. The receiver shall bear the ASME stamp and no welding or other activation shall be permitted after the receiver has been certified. Unless otherwise specified (see 6.2), the capacity of the air-receiver tanks for the 5-cfm compressors shall be not less than 60 gallons and for the 15-cfm and 25-cfm compressors shall be not less than 80 gallons. Unless otherwise specified (see 6.2),the air-receiver tank fittings shall include a globe discharge valve, a drain cock, and a 0 to 300-psig pressure gage having 10 psig or smaller increments. The air-receiver tanks shall include a pressure relief valve conforming to ASME Boiler and Pressure Vessel Code. The receiver shall be provided with a steel base or not less than four mounting feet for horizontal mounting. When equipped with a steel base, openings shall be provided to permit insertion of forklift tines. Each opening shall be not less than 11 inches by 3 inches in size; openings shall be located such that the center of gravity of the compressor falls between the centers of forklift openings. A means shall be provided on the receiver for securing the airhose.

3.22 Class 2, handtruck mounted. The compressor unit and power unit shall be mounted on a lightweight, 2-wheel handtruck braced to minimize vibration. A handle shall be provided which will permit carting by one man when the handle is elevated to a height of not less than 30 inches and not more than 48 inches. The class 2 compressor shall be equipped with a combination storage rack and landing legs mounted on the handle end of the truck similar to that shown on figure 1 (see 6.8). The landing legs shall be equipped with soft rubber feet so spaced as to minimize movement of the compressor during operation. The complete assemblage shall withstand vibration, falling, and bumping, encountered by this type of equipment without malfunction or damage when operated as specified herein. The dry weight of the class 2, size 5-cfm compressor shall not exceed 150 pounds (see 6.13).

3.22.1 Air-receiver tank. The air-receiver tank for the class 2 compressor shall have a capacity of not less than 2 gallons and shall be constructed in accordance with the ASME Boiler and Pressure Vessel Code, Section VIII. Air-receiver tanks with an inside diameter of 6 inches or less, which are not within jurisdiction of the ASME Boiler and Pressure Vessel Code, Section VIII, shall be designed for a proof pressure of 300 psig. Unless otherwise specified (see 6.2), the air-receiver tank fittings shall include a drain cock, and a 0- to

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300-psig pressure gage having 10 psig or smaller increments. The air receiver tanks shall include a pressure relief valve conforming to ASME Boiler Pressure Vessel Code. The pressure gage shall be mounted in a manner to prevent damage due to vibrations.

3.22.2 Wheels and hubs. The wheels and hubs for class 2 compressor shall be of the drop center type to accommodate the tires specified in 3.22.3. The wheels and hubs shall be furnished with roller bearings and seals.

3.22.3 Tires and tubes. The tires for the class 2 compressor shall conform to ZZ-T-410, size 4.10.6. The tires shall be 2-ply rating. Tubes shall conform to ZZ-I-550.

3.22.4 Tilt requirement. The compressor shall deliver rated capacity when operated at rated speed and pressure when tilted 15 degrees from the horizontal in any plane.

3.23 Class 3, caster-mounted. The compressor unit and power unit shall be mounted on a steel base attached to a horizontal-type, air-receiver tank. The air-receiver tank shall be constructed as specified in 3.21.1, except the tank shall be mounted on four casters conforming to FF-C-88. Two type II (swivel) casters shall be mounted on the compressor end, and two type I (nonswivel) casters shall be mounted on the engine end.

3.23.1 Caster installation. The casters shall be mounted so that the compressor will not overturn when pushed or pulled about on a horizontal surface or when tilted 15 degrees from the horizontal in any direction. The casters shall be attached to steel mounts on the bottom side of the receiver tank and located near each end of the tank. The distance between the centerline of the casters (measured side to side) shall be not less than 6 inches greater than the maximum diameter of the air-receiver tank.

3.23.2 Handle and storage rack. The class 3 compressor shall be equipped with a combination handle and storage rack mounted on the compressor end. The handle and storage rack shall be removable, and similar to that shown on figure 2 (see 6.8). The handle shall be positioned not less than 30 inches and not more than 40 inches from the floor. The storage rack shall hold the air hose specified in 3.9. The storage rack shall be located so that the hose when stored on the rack, will not contact any component of the compressor subject to high temperatures.

3.24 Class 4, trailer mounted. The class 4 air compresor shall be mounted on a trailer as specified in MIL-C-52437, class CCE, size 3. The CCE trailer chassis shall be equipped with a military compatible 12 and 24 volt system as specified. The engine compressor unit and receiver shall each be directly assembled on a common base.

3.24.1 Cover. A canvas cover for storage and protection during towing shall be provided. The materials, stitching, thread, and seam of the canvas cover shall conform to MIL-C-13489. The canvas cover shall be three dimensional to enclose the engine and compressor and shall be fastened by means of a line drawn through grommets in the seam or hem.

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3.24.2 Toolbox. A toolbox made of not less than No. 14 gage sheet steel shall be furnished with the lid edges formed to overlap the top edges of the box to prevent the entry of dirt, snow, and rain. The toolbox shall be of sufficient size to hold the canvas cover, special tools, inflator gage and publications.

3.24.3 Cranking access. Cranking shall be accomplished manually by means of a rope. The engine shall be mounted in relation to the compressor unit so that the starter pulley is accessible with clearance for starting.

3.24.4 Storage rack. A storage rack shall be furnished and installed on each compressor for wrapping or looping the air hose when not in use. The rack shall be installed on the compressor so that the hose will not come in contact with any component of the compressor subject to high temperature or abrasion.

3.24.5 Air-receiver tank. The capacity of the air-receiver tank shall be not less than 20 gallons and shall be constructed in accordance with the ASME Boiler and Pressure Vessel Code, Section VIII. The receiver shall bear the ASME stamp and no welding or other activation shall be permitted after the receiver has been certified. Unless otherwise specified (see 6.2), the air-receiver tank fittings shall include a spring-loaded pressure relief valve, a globe discharge valve, a drain cock, and a 0- to 300-psig pressure gage having 10 psig or smaller increments. The air-receiver tanks shall include a pressure relief valve conforming to ASME Boiler and Pressure Vessel Code.

3.24.6 Lifting attachments. The base-mounted 15-cfm compressor shall be equipped with permanently affixed lifting attachments conforming to MIL-STD-209 that will enable the compressor to be lifted in its normal travel position. Instructions shall be stenciled above or near each attachment warning that the lifting attachment shall not be used when the compressor is mounted on the trailer chassis.

3.24.7 Mobility. The class 4, size 15-cfm compressor, chassis combination shall be capable of being towed over paved roads, unimproved surfaces and cross-country terrain at normal towing speeds.

3.25 Class 4, size 15-cfm, winterized. When specified (see 6.2), the compressor shall be winterized for operation at minus 50° F. When the winterized compressor is specified, the canvas cover, as specified in 3.24.1, shall not be furnished. The winterized compressor shall be provided with the following items in addition to those specified in 3.24.

3.25.1 Enclosure. A metal enclosure shall be provided to completely enclose the top, bottom, sides, and ends of the compressor assembly, to prevent the entry of blowing snow, rain, and dust, and to maintain temperature while heating and operating the compressor at minus 50° F. Provisions shall be made for opening side panels or doors and manually adjustable louvers at the engine end to allow ventilation for the engine and compressor unit at 120° F without excessive overheating as specified in MIL-STD-1300 for the engine, and exceed the temperature ratings as specified by the compressor manufacture for the

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compressor unit. Weatherstripping shall be provided on all doors and panels. Weatherstripping shall have a protective wear coating, and both the weatherstripping and protective wear coating shall be flexible at minus 50° F. The weatherstripping shall conform to MIL-R-14328. The weatherstripping shall not interfere with normal operation. Insulation shall be furnished on the interior of the enclosure and shall be of the mastic type and not less than 3/32 inch thick.

3.25.2 Heater duct connection. A heater duct connection shall be provided on the enclosure for connection of a 12-inch-diameter duct from a heater conforming to MIL-H-4607, class 3. The duct connection and cover shall conform to MIL-H-7365, size A. The duct connection cover shall be securely fastened to the enclosure by a keeperchain or other means to prevent loss when removed. The heater duct connection shall not protrude beyond outer extremes of the enclosure more than necessary to facilitate connection of th heater duct.

3.25.2.1 Ducting. Provisions shall be made for adequate ventilation and circulation of air for the compressor, while preheating and operating under cold weather conditions. The heat from the heater shall flow through the compartment in such a manner as to warm the entire engine and compressor unit to permit starting of the compressor in not more than 30 minutes at minus 50° F. Ducting or baffles shall be provided to divert heated air to the batteries and to all critical moving parts which may be restricted due to ice or congealed oil or grease. The heat shall be exhausted from the compartment through a manually operated louver in such a manner as to flow over the receiver, hose rack, and hose, warming these parts prior to starting.

3.25.3 Doors, panels, and openings. Doors or panels shall be furnished with recessed latches operable by personnel wearing heavy arctic mittens. The doors or side panels shall be of a size and location to afford accessibility to the compressor unit, control equipment, and engine for routine service, adjustment, and maintenance. Doors over 18 inches in vertical height shall be hinged horizontally in two sections. Hinges shall be the continuous piano type with brass hinge pins. The roof of the enclosure shall be removable so that installed components can be removed or replaced vertically.

3.25.4 Engine. The Government-furnished engine (see 3.17 and 6.5) will include an electric and manual starting system and a 5-ampere charging generator as specified in MIL-STD-1300. The Government-furnished starter (see 3.17 and 6.5) shall be installed as shown in drawing 13208E5819. All parts required for mounting shall be furnished by the contractor.

3.25.4.1 Engine (commercial). The commercial engine (see 3.26) shall include an electric manual starting system and an appropriate charging system of not less than 5 ampers.

3.25.5 Batteries. Storage batteries and battery box shall be furnished in accordance with MIL-STD-1410.

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3.25.6 Fuel tank. A fuel tank as specified in 3.26.3 shall be provided, except that the tank shall be sufficient capacity for 8 hours operation at rated capacity.

3.25.7 Belts. All drive belts and fan belts shall conform to MIL-B-11040.

3.25.8 Control devices. The engine shall be provided with a high cylinder-head temperature device, set to automatically stop the engine in the event of excessive cylinder-head temperatures. The control device shall be set so that the head temperature specified in MIL-STD-1300 will not be exceeded.

3.25.9 Control panel. A control panel shall be provided for the class 4, size 15-cfm, winterized compressor with all compressor gages, meters, valves, switches, and other operator controls as specified herein. The control panel shall be located within the enclosure, shall be shock-mounted, and shall be accessible through a watertight, hinged cover of sufficient size to expose all controls and instruments when open. Opening of the control panel cover shall not vent the engine and compressor unit compartment. Each instrument, meter and control device shall be identified by a nameplate or by legible lettering on its face. All gages and meters shall be uniformly flush mounted. A schematic airflow diagram, operating instructions, and safety precautions shall be attached to the panel or control panel cover, so as to be visible to the operator when the control panel is in the open position. The control panel shall include the following equipment:

a. Oil pressure gage (for engine having forced-feed lubrication above 15 psig).
b. Ammeter.
c. Fuel gage, liquid level.
d. Engine hourmeter.
e. Starter and ignition switches.
f. Throttle and choke, or governor controls.
g. Temperature gage, engine oil, indicating.
h. Pressure gage, discharge air pressure (psig).
i. Pressure gage, compressor lubricating oil pressure (psig) (if forced feed lubricated).

3.25.10 Lifting attachments. Lifting attachments shall be furnished as specified in 3.24.6.

3.25.11 Fuels and lubricants. When specified (see 6.2), the winterized class 4, size 15-cfm compressor shall be serviced prior to shipment with military lubricants and greases as specified in table II:

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TABLE II. Service products.

Product	Specification	Type and Grade
Gasoline, auto, combat, 91 octane	MIL-G-3056	Type II
Lubricating oil, internal combustion engine, arctic	MIL-L-46167	OES
Greases:		
Aircraft and instrument	MIL-G-23827	GL
Automotive and artillery	MIL-G-10924	GAA

3.26 Gasoline-engine. Unless otherwise specified (see 6.2), the engine shall be Government furnished. When contractor furnished commercial engines are used, they shall have a cast iron cylinder liner and stellite exhaust valves and seats.

3.26.1 Government-furnished engine. A Government-furnished military design engine will be provided for the 5-cfm, and 15-cfm compressors (see 3.17). The contractor shall install the engine in accordance with MIL-STD-1226 for the 1-1/2-bhp military design engine and MIL-STD-1300 for the 6-bhp military design engine. When a multiple V-belt drive is used, the engine power takeoff shall be mounted as specified in MIL-STD-1226 or MIL-STD-1300, as applicable.

3.26.1.1 Power and speed. The size 5-cfm compressor shall require not more than 1.5 hp when delivering 5-cfm, 175 psig air at an engine speed of 3600 rpm. The 15-cfm compressor shall require not more than 6 hp when delivering 15-cfm, 175 psig air at an engine speed of 3600 rpm.

3.26.2 Commercial engine. For the purpose of selecting the engine, the compressor shall be classified as a class II end item in accordance with MIL-STD-1410. The engine, including all systems, components, and accessories, shall conform to MIL-STD-1410, except as otherwise specified herein. The engine shall be 4-cycle and air cooled. The engine shall be furnished as a complete, self-contained, power package including hand throttle control, fail-safe governor, manual starting system and engine stop switch. If automatic rewind manual starter is used, it shall have in addition, a pulley for manual rope starter to be used in event of failure of the automatic rewind. The engine shall be capable of operating under all conditions specified herein using fuels and lubricants specified herein and in MIL-STD-1410, without exceeding any of the operating limits specified herein in table I of MIL-STD-1410, or the engine manufacturer's recommended operating limits.

3.26.2.1 Size 25 CFM compressor. The engine shall be furnished with a 12-volt cranking and battery charging system in accordance with MIL-STD-1410. The battery charging alternator shall have a capacity of not less than 15 ampere capacity. A flywheel type alternator is acceptable. A dry type air cleaner with a restriction indicator (to warn the operator when the air cleaner requires servicing) shall be furnished in accordance with MIL-STD-1410. Standard commercial batteries shall be furnished.

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3.26.3 Fuel tank. Except as specified in 3.25.6, a fuel tank with a capacity for not less than 4 hours of operation at rated load shall be provided. Construction and mounting shall be such that vibration, tightening of the mounting devices, or contact with other components of the engine or compressor will not cause leaks to develop. When a fuel tank of 10 gallon capacity or more is furnished, a filler neck assembly conforming to MS35644 shall be provided and shall incorporate a removable strainer, and have a chain secured filler cap and a fuel tank cap gasket. When a fuel tank of less than 10 gallon capacity is furnished, the filler opening shall be a minimum of 2.250 inches and shall incorporate a removable strainer and a chain-secured filler cap. The filler cap shall be a screw cap that will prevent fuel from leaking when the compressor is tilted 15 degrees from the horizontal in any plane and when the handtruck-mounted compressor is elevated for mobility. The tank shall be provided with an air vent and a drain. A standpipe at least 1/4-inch above the bottom shall be incorporated in the fuel tank outlet to prevent the entry of dirt and foreign matter into the system. The tank and all lines shall be located or positioned to minimize fire hazard or damage resulting from spilling, overflow, or drainage. Provision shall be made for connection to an auxiliary fuel supply. The fuel tank shall be equipped with a fuel level device.

3.26.4 Engine pulley. When a V-belt drive is used, the engine drive pulley shall be provided with an inner hub which clamps to the shaft by moving axially along a taper in the outer hub. The mechanical clamping shall be accomplished by the use of a screw member or members for mounting and removal of the pulley.

3.26.5 Exhaust system. The muffler for the 1-1/2-hp engine will be furnished by the Government as part of the engine. Two mufflers shall be furnished by the contractor for the 6-hp engine. The back pressure of the exhaust system, including mufflers, shall not exceed 8 inches of water at maximum horsepower output and speed under all operating conditions specified herein. The exhaust manifold and mufflers shall be positioned so that exhaust gases will be directed away from the fuel tank, compressor air intake, and any other component or function that may be adversely affected by exhaust gases or heat and shall remain at least 4 inches away from the fuel tank under all belt adjustments. The exhaust system shall be protected against the entry of rain and shall include drainage provisions to prevent accumulation of water and condensed vapors.

3.26.6 Power takeoff. When a V-belt drive is used for the size 15-cfm and 25-cfm compressors, the contractor shall furnish and install a power takeoff on the engine for driving the compressor unit. The power takeoff shall consist essentially of a 4-bolt flange bolted to the engine flywheel and a stub shaft or other device on which the drive pulley or clutch shall be mounted. When a direct drive is used, a flexible coupling shall be installed between the engine and compressor units. The nominal rating of the power takeoff shall be not less than 150 percent of the rated horsepower of the engine.

3.26.7 Clutch. When specified for the 15- and 25-cfm size compressors (see 6.2), the engine shall be equipped with a clutch. The clutch shall be accessible for adjustment, replacement of parts, and lubrication (unless

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bearings with sealed-in lubrication are used). Adjustment of the clutch shall not require special tools. The nominal rating of the clutch shall be not less than 150 percent of the maximum engine torque. The clutch operating lever shall be accessible to the operator.

3.26.8 Pressure regulation. The compressor shall have an adjustable, pressure actuated mechanical control of the head unloading type. The control shall be adjusted to stop compression of air when the tank pressure has risen to 200 psig, plus 0 psig, minus 10 psig and to start compression of air when the tank pressure has dropped to 175 psig, plus or minus 10 psig. Means shall be provided to prevent the engine from starting against tank pressure or cylinder head pressure.

3.26.8.1 Size 5, class 2. The size 5, class 2 compressor may utilize a pop valve type unloader.

3.26.8.2 Size 15 and 25. The size 15 and 25 compressor shall have a device to automatically reduce the engine speed to idle speed when the compressor is unloaded. A manual control shall be provided to by-pass the engine idle device.

3.27 Interchangeability. All parts having the same part number shall be functionally and dimensionally interchangeable. Interchangeable parts are defined as two or more parts possessing such functional and physical characteristics as to be equivalent in performance and durability and capable of being exchanged one for the other without alteration of the parts themselves or of adjoining parts, except for adjustment, and without selection for fits or performance.

3.28 Workmanship. All parts, components and assemblies of the compressor including castings, forgings, molded parts, stampings, bearings, seals, machined surfaces, and weld parts shall be clean and free from sand, dirt, fins, pits, sprues, scales, or any damaging extraneous material. The compressor shall be free of any defect that could impair its operation and serviceability.

3.28.1 Fabrication. Steel and aluminum used in the fabrication of the compressors shall provide original quality surface finish and shall be free from kinks and sharp bends. The forming of the material shall be done by methods that will not cause damage to the metal. Shearing and chipping shall be done neatly and accurately. Corners shall be square and true, and sharp edges and burrs shall be removed. Flame-cutting may be employed instead of shearing and sawing. Burned surfaces of flame-cut material shall be ground or machined sufficiently to remove ash and cooling checks. Precautions shall be taken to avoid overheating, and heated metals should be allowed to cool slowly. All bends of a major character shall be made with metal dies or fixtures to insure uniformity of size and shape.

3.28.2 Welding. The surfaces of parts being welded shall be free of scale, paint, grease, and other foreign matter. Welds shall transmit stress without permanent deformation or failure when the parts connected by the welds are subjected to proof and service loading.

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3.28.3 Bolted connections. Boltholes shall be accurately punched or drilled and shall have the burrs removed. Boltholes in structural members subject to variable loads shall be punched or drilled. Washers or lockwashers shall be provided where necessary. All bolts, nuts, and screws shall be correctly torqued. Matching thread areas securing bolts or capscrews shall be of sufficient strength to withstand the tensile strength of the bolt.

3.28.4 Riveted connections. Rivets shall fill the holes completely. The upset rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member and shall be in accordance with SAE J492.

3.28.5 Machine work. Tolerances and gages for metal fits shall conform to the limitation specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined and tested in accordance with referenced specifications and standards.

4.2 Classificaion of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection comparison (see 4.6).
d. Inspection of packaging (see 4.7).

4.3 First article inspection.

4.3.1 Examination. The first article compressor shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The first article compressor shall be tested as specified in 4.5.2.1 through 4.5.2.21 as applicable for the class compressor to be furnished. Failure of any test shall be cause for rejection. A first article test log (see figure 3) shall be furnished.

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4.4 Quality conformance inspection.

4.4.1 Examination. Each compressor shall be examined in accordance with 4.5.1.

4.4.2 Tests.

4.4.2.1 Individual. Each compressor shall be tested as specified in 4.5.2.2 and 4.5.2.22. Failure of a test shall be cause for rejection.

4.5 Inspection procedure.

4.5.1 Examination. The compressor shall be examined for the following defects:

101. Material not as specified.
102. Materials not resistant to corrosion and deterioration or treated to be made resistant to corrosion and deterioration.
103. Dissimilar metals not in accordance with MIL-STD-889.
104. Material and material finish or treatment information not available.
105. Recovered materials not as specified.
106. Human factors not as specified.
107. Safety devices or guards missing or not as specified.
108. Ease of maintenance not as specified.
109. Airhose not as specified.
110. Intake air not as specified.
111. Lubrication not as specified.
112. Lubricants not as specified.
113. Lubrication fittings not as specified.
114. Pressure-release device not as specified.
115. Enclosures with integral reservoir or lubricant not as specified.
116. Delivery lubrication not as specified.
117. Fungus and moisture resistance not as specified.
118. Identification marking and instruction plates not as specified.
119. Treatment and painting not as specified.
120. V-belts and pulleys not as specified.
121. Clutch not as specified.
122. Air receiver tank not as specified.
123. Base-mounted compressor not as specified.
124. Handtruck mounted compressor not as specified.
125. Installation of Government furnished engine not as specified.
126. Wheels and hubs not as specified.
127. Tires and tubes not as specified.
128. Handle and storage rack not as specified.
129. Cover not as specified.
130. Toolbox not as specified.
131. Cranking access not as specified.
132. Storage rack not as specified.
133. Enclosure not as specified.
134. Heater duct connection not as specified.

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135. Ducting not as specified.
136. Doors, panels and openings not as specified.
137. Fuel tank not as specified.
138. Control panel not as specified.
139. Fuels and lubricants not as specified.
140. Interchangeability not as specified.
141. Assembly incomplete, components missing or not as specified.
142. Workmanship not as specified.
143. Parts, special tools, and items to be furnished with compressor missing or not as specified.
144. Fabrication not as specified.
145. Welding not as specified.
146. Bolted connections not as specified.
147. Riveted connections not as specified.
148. Machine work not as specified.
149. Pressure tubing or fitting not rated as specified.

4.5.2 Tests.

4.5.2.1 Test conditions. Unless otherwise specified in a test, tests shall be performed without shelter and at the climatic conditions existing at the place of test. The compressor shall operate as specified herein without maintenance other than the contractor's recommended normal scheduled maintenance as established by a maintenance schedule prepared and submitted by the contractor prior to test. The compressor shall be lubricated with military oils and greases. Oils shall be those designated for use at the ambient temperature at the place of test. A first article test log similar to figure 3 shall be furnished tabulating all appropriate test data and observations. Safety and human factor considerations shall be evaluated throughout testing.

4.5.2.2 Air receiver tank. The air receiver tank for the compressor shall be tested in accordance with the ASME Boiler and Pressure Vessel Code, Section VIII. Failure of any ASME tests shall constitute failure of this test. Presence of the ASME official code U-symbol stamped or marked on the receiver and the furnishing of a copy of the ASME data sheet (Form U-1) may be accepted as evidence that the receiver conforms to the ASME Boiler and Pressure Vessel Code. For the class 2 compressor, when furnished with an air receiver tank with an inside diameter of 6 inches or less, which are not within the jurisdiction of the ASME Boiler and Pressure Vessel Code, expose the air receiver tank, exclusive of pressure gages plus safety relief devices, to a hydrostatic pressure of 300 psig and hold at this pressure until examined for leaks and distortion. Any evidence of leakage or permanent distortion shall constitute failure of this test.

4.5.2.3 Power consumption. Drive the compressor by a dynamometer or calibrated motor and determine the brake horsepower and speed required to deliver minimum capacity at maximum operating pressure as specified in table I. The occurance of one of the following shall constitute failure of this test:

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a. Size 5-cfm compressor, with Government-furnished engine, requiring more than 1.5 continuous brake horsepower at an engine speed of 3600 rpm.
b. Size 15-cfm compressor, with Government-furnished engine, requiring more than 6 continuous brake horsepower at an engine speed of 3600 rpm.
c. Size 5, 15 and 25-cfm compressors for commercial engines, horsepower and speed required for operation of compressor in excess of the continuous brake horsepower rating for applicable speed as recommended by the engine manufacturer.

4.5.2.4 Capacity and performance. The compressor capacity shall be determined in accordance with the method specified in the ASME Performance Test Code 9 - Displacment Compressors, Vacuum Pumps, and Blowers. This test shall be performed in conjunction with and simultaneously with the endurance test (see 4.5.2.5). The air-discharge temperature shall be determined when the compressor is operating at minimum capacity at maximum discharge pressure. Any one of the following conditions shall constitute failure of this test:

a. Discharge air pressure less than 175 psig.
b. Discharge air temperature in excess of ambient plus 100° F.
c. Free-air capacity less than that specified in table I.

4.5.2.5 Endurance. Operate the compressor for not less than 500 hours continually in periods of not less than 8 hours of running time. For this test, an auxiliary fuel supply system may be used to permit operation for 8 hours without refueling. If an auxiliary fuel system is not used, periods of nonoperation of not more than 10 minute duration will be allowed for refueling. The refueling time shall not be counted as operating time but shall not cause disallowance of the operating period during which it takes place. Oil changes and other preventive maintenance shall be performed during downtime. There shall be no replacment of parts. Adjustments provided by integral devices and repairs which require 1 man-hour or less may be made. The 500 hours of operation shall be accomplished at rated capacity and pressure. Record appropriate remarks, including a record of all stops and adjustments. Record the following data at least once during each 24 hour period. Readings shall be taken only after at least 1 hour of continuous operation:

a. Air temperature, ambient.
b. Air temperature at receiver discharge.
c. Air receiver pressure.
d. Barometric pressure.
e. Air temperature at compressor intake.
f. Air temperature upstream of nozzle.
g. Air pressure of compressor intake.
h. Nozzle throat diameter.
i. Nozzle coefficient of discharge.
j. Differential pressure across the nozzle.
k. Engine speed (rpm).
l. Compressor unit speed (rpm).

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m. Compressor capacity (cfm) computed.
n. Engine intake air temperature.
o. Engine oil pressure (if pressure lubricated).
p. Engine oil sump temperature.

Completion of less than 500 hours of operation at rated capacity and pressure, discharge air temperature in excess of ambient plus 100° F., requirement for replacement of parts, requirement for repairs requiring more than 1 man-hour to complete, or necessity for adjustment not provided for by integral devices shall constitute failure of this test.

4.5.2.6 High temperature. Unless otherwise specified (see 4.5.2.18), subject the compressor to an ambient temperature of 120° F. plus or minus 5° F. until all components have stabilized at this temperature. Then start and operate the compressor at rated capacity and pressure for not less than 4 hours while maintaining 120° F. plus or minus 5° F. ambient temperature. Adjustments to the engine or compressor unit shall not be permitted after the first hour of operation. The operating time for this test may be counted toward the accomplishment of the endurance test. Record the data as specified in 4.5.2.5 at 30-minute intervals. Failure to start and operate at rated capacity, pressure and speed, discharge air temperature in excess of ambient plus 100° F. or requirement for replacement of parts shall constitute failure of this test.

4.5.2.7 Low temperature. Unless otherwise specified (see 4.5.2.19), subject the compressor to an ambient temperature of minus 25° F. plus or minus 5° F. until all components have stabilized at this temperature. Then start and operate the compressor at rated capacity, pressure and speed for not less than 4 hours while maintaining minus 25° F. plus or minus 5° F. ambient temperature. Adjustments to the engine or compressor unit shall not be permitted after the first hour of operation. The operating time for this test may be counted toward the accomplishment of the endurance test. Record the data as specified in 4.5.2.5 at 30-minute intervals. Failure to operate at rated capacity, pressure and speed, or requirement for replacement of parts shall constitute failure of this test.

4.5.2.8 Cycling. Start the compressor and allow the receiver pressure to build up until the pressure-regulation control operates. Then reduce the receiver pressure until the pressure-regulation control activates to increase the pressure. Vary the pressure as above until a total of 20 high-pressure actuations and 20 low-pressure actuations have been reached. Inability of the pressure-regulation control to control the pressure as specified shall constitute failure of this test.

4.5.2.9 Pressure regulation, size 5-cfm. When an adjustable pressure-regulating valve of the pressure-relief type actuated by receiver pressure is used, at the beginning and after completion of the endurance test, start the compressor and allow the receiver pressure to build up to determine pressure required to discharge full capacity through the pressure-regulating valve. Then reduce the receiver pressure to determine the pressure at which the pressure-regulating valve reseats. Inability to meet the requirements specified in 3.26.8 shall constitute failure of this test.

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4.5.2.10 Electromagnetic interference. The compressor shall be tested in accordance with MIL-STD-462 to determine conformance to 3.11. The contractor shall furnish the contracting officer the first article test report for evaluation. Failure to meet the requirments of 3.11 shall constitute failure of this test.

4.5.2.11 Sound level. Place the compressor in a test area which is free of any large reflecting surfaces within 50 feet of the compressor or sound-level measuring equipment. With the compressor operated at rated load, a sound level survey shall be made around the compressor to determine the direction in which the maximum sound levels are propagated so that the test equipment can be properly oriented. The recording microphones shall be positioned in accordance with MIL-STD-1474, category D. The noise level shall be measured with the compressor operated at no load and rated load. Failure to meet the requirements of 3.7.3 shall constitute failure of this test.

4.5.2.11.1 Exterior noise. The compressor shall be tested in accordance with EPA 40 CFR 204 for portable air compressors. Nonconformance to 3.7.3.3 shall constitute failure of this test.

4.5.2.12 Class 2.

4.5.2.12.1 Stability. During the performance of the endurance test, measure the movement during three 8-hour periods of operation. The first period shall be between 0 to 25 hours, the second period between 250 and 300 hours, and the third, between 450 and 500 hours. At the beginning of each of these 8-hour periods, place the landing feet of the handtruck within circles marked, scribed, etched, or painted on the floor. The circles shall be 6 inches in diameter or 2 inches larger in diameter than the rubber feet, whichever is larger. The compressor shall be held to the floor by gravity alone, and movement shall in no way be restricted by mechanical attachments, braces, wheel chocks, or other means integral to the compressor. Location of either or both of the rubber feet, wholly or in part, outside of or upon the circles at any time during the 8-hour period of operation shall constitute failure of this test.

4.5.2.12.2 Mobility. Prior to beginning the last 250 hours of operation of the endurance test, stop the compressor and test for mobility as follows:

a. Elevate the handle to a height of not less than 30 inches nor more than 48 inches and move the compressor not less than 100 yards turning in complete circles to the right and left during the movement. At least 50 yards of the movement shall be in a straight line. Cart the compressor over a 12-inch drop onto a concrete surface not less than three times.
b. Tilt the compressor to an angle of not less than 45 degrees by elevating one side of the frame and allowing the wheel and landing leg on the opposite side of the frame to rest on a concrete floor. Allow the elevated side to fall to a position of rest on the concrete floor without any restraining forces being applied. Do this not less than two times for each side.

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c. Inability to be moved by one man, inability to be carted in a straight path, inability to be turned either to the right or the left, or fracture or permanent deformation of any component or part or leakage of gasoline from the tank shall constitute failure of this test.

4.5.2.12.3 Tilted operation. Tilt the compressor 15 degrees from the horizontal about the longitudinal axis and operate in this position for 30 minutes. This test shall be performed in each of the four directions (compressor tilted with respect to longitudinal and transverse axes). Allow the compressor to stand for 4 hours in the tilted position most likely to adversely affect the lubrication or fuel system, then start and operate the compressor for 30 minutes. Any malfunction due to tilted operation, permanent deformation of any part of component, or leakage of gasoline from the tank shall constitute failure of this test.

4.5.2.13 Class 3.

4.5.2.13.1 Casters. Push and pull the compressor horizontally in all directions, turn the compressor 360 degrees, and tilt the compressor in 15 degrees from the horizontal in all directions. Overturning or any condition that interferes with the maneuverability of the compressor or physical damage to the compressor shall constitute failure of this test.

4.5.2.14 Exhaust and fuel system.

4.5.2.14.1 Engine exhaust. Measure the engine exhaust pressure from a pressure tap approximately 2 inches from the exhaust manifold flange with the compressor operating at the maximum speed and the horsepower output specified herein. Exhaust pressure in excess of that specified in 3.26.5 shall constitute failure of this test. When the engine exhaust muffler is furnished by the Government as part of the military design engine, this test is not required.

4.5.2.14.2 Gasoline tank capacity. Fill the gasoline tank. Start the compressor and operate at rated capacity and pressure until the engine ceases to operate due to lack of fuel. The elapsed time of operation shall be recorded. No refueling or auxiliary fuel supply shall be allowed. Elasped operation time of less than as specified in 3.26.3 or 3.25.6 shall constitute failure of this test.

4.5.2.15 Road test. Tow the class 4, size 15-cfm compressor, for a distance of 100 miles at maximumn safe speeds. Twenty-five miles of this travel shall be on cross-country terrain (see 6.9) and 75 miles shall be on primary roads. Operation on smooth, paved surfaces shall include not less than 15 abrupt stops from a speed of 25 mph in the forward direction, not less than five abrupt stops from a speed of 2 mph in the reverse direction, and not less than 10 sharp turns to the left and 10 to the right at not less than 25 mph. Radii of the sharp turns shall be not greater than 150 feet. Displacement or damage to welded, bolted, or riveted connections or failure to operate as specified shall constitute failure of this test.

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4.5.2.16 Center of gravity test. Determine the horizontal center of gravity of the class 4, size 15-cfm compressors when mounted on the trailer as specified in 3.24. Nonconformance to 3.24 shall constitute failure of this test.

4.5.2.17 Tilted operation - class 4. Tilt the class 4, size 15-cfm compressor 15 degrees from the horizontal about the longitudinal axis and operate in that position for 30 minutes. Repeat this test in each of the four directions (compressor tilted with respect to the longitudinal and transverse axes). Allow the compressor to stand for 4 hours in the tilted position most likely to adversely affect the lubrication or fuel system and then start and operate the compressor for 30 minutes in that position. Any malfunction due to tilted operation such as lack of lubrication, rough or erratic operation, or permanent deformation of any part or component shall constitute failure of this test.

4.5.2.18 High temperature. Conduct the high temperature test on the class 4, size 15-cfm compressor as specified in 4.5.2.6. The test shall be performed with the canvas cover removed on the nonwinterized compressor and with all side panels, louvers, and the duct in the open position or removed on the winterized compressor. In addition to the data specified in 4.5.2.5, record the compressor unit and engine lubricating oil sump temperatures, at 30-minute intervals. Failure to start and operate at rated capacity, pressure, and speed, oil temperature higher than specified in MIL-STD-1300 for the engine and as specified by the contractor for the compressor unit, discharged air temperature in excess of ambient plus 100° F., or requirement for replacement of parts shall constitute failure of this test.

4.5.2.19 Low temperature. Subject the class 4, size 15-cfm winterized compressor to an ambient temperature of minus 50° F. until the batteries and the engine and compressor unit lubricating oil temperatures have stabilized. A 30-minute preheat with a heater conforming to MIL-H-4607, class 3, or equivalent, shall be utilized prior to attempting to start at minus 50° F. Start the compressor with the batteries that were cold soaked with the compressor. Outside power for cranking shall not be used. After starting, disconnect the heater, replace the duct cover, and operate the compressor at rated speed, capacity and pressure for not less than 4 hours. Record the data as specified in 4.5.2.5 at 30-minute intervals. Failure to start after the 30-minute preheat period, failure to operate at rated speed, pressure and capacity after starting, or requirment for replacement of parts, shall constitute failure of this test.

4.5.2.20 High temperature shutoff. On the winterized class 4, size 15-cfm compressor, activate the high cylinder-head temperature control device to stop the engine to determine conformance to 3.25.8. The device may be activated after complete or partial disassembly from the engine and by the applicaton of an outside high temperature source to simulate high cylinder-head temperature.

4.5.2.21 Lifting attachment. Lift class 4, size 15-cfm compressor in its normal travel position using only the permanently affixed lifting attachments. Determine the force and angle of application for each attachment and clearance

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between each sling and the compressor. Then anchor the compressor and subject each attachment to a load equal to 2-1/2 times the load determined above and applied at the angle determined above. Apply each load for not less than 90 seconds. Weld failure or permanent deformation of the lifting attachments or structural members shall constitute failure of this test.

4.5.2.22 Capacity. Test each compressor for capacity in accordance with 4.5.2.4 except that each compressor shall be tested as a unit driven by its gasoline engine. Prior to testing, run-in each compressor as prescribed by the contractor. Occurance of one or more of the following conditions shall constitute failure of this test:

a. Discharge air pressure less than 175 psig.
b. Discharge air temperature in excess of ambient temperature plus 100° F.
c. Free-air capacity less than that specified in table I.

4.6 Inspection comparison. The Government may select compressors at any time during the contract production period and subject these compressors to the examination specified in 4.5.1 and tests specified in 4.5.2 to determine conformance to the requirements of this specification. The inspection will be performed by the Government, at a site selected by the Government, on units selected at random from those which have been accepted by the Government and will not include the previously inspected first article model compressors. In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison, and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

4.6.1 Inspection failure. Failure of an inspection comparison compressor to meet any requirement specified herein during and as a result of the examination and tests specified in 4.6 shall be cause for rejection of the inspection comparison compressor(s) and shall be cause for refusal by the Government to continue acceptance of production compressors until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on compressors previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all compressors accepted prior to the completion of inspection comparison are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.7 Inspection of packaging. The preservation, packing, and marking shall be examined and tested to determine compliance with the quality assurance provisions of MIL-C-3600.

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5. PACKAGING

5.1 Preservation, packing and marking. The air compressors, accessories, repair parts, tools, and publications shall be preserved, packed and marked in accordance with MIL-C-3600. The level of preservation and the level of packing shall be level A, B or commercial as specified (see 6.2).

6. NOTES

6.1 Intended use. The compressors are intended for use as a source of compressed air in the normal operations conducted at filling stations and vehicle maintenance shops.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Type, class, and size required (see 1.2).
c. Time frame required for submission of first article model and number of compressors to be furnished (see 3.2).
d. When the Government will conduct any or all of the first article examination and tests. When the Government will conduct some but not all of the first article examination and tests, the contracting officer should specify which examination and tests will be conducted by the Government and which examination and tests shall be conducted by the contractor (see 3.2 and 3.3).
e. When winterized first article examination and testing is required and number of compressors to be furnished (see 3.3).
f. Length of air hose to be furnished if other than as specified (see 3.9).
g. When electromagnetic interference requirements other than as specified in 3.11 are required.
h. Color required (see 3.16).
i. When Government-loaned property other than as specified in 3.18 is required.
j. Capacity of air receiver tank if other than as specified (see 3.21.1).
k. Air-receiver-tank fittings required if other than as specified (see 3.21.1, 3.22.1, and 3.24.5).
l. When winterized class 4, size 15-cfm compressors are required (see 3.25)
m. When winterized class 4, size 15-cfm compressors will be serviced prior to shipment (see 3.25.11).
n. When sizes 15-cfm and 25-cfm compressors will be furnished with a clutch (see 3.26.7).
o. Degree of preservation and packing required (see 5.1).
p. When commercial engines are to be used for size 5 and 15 cfm compressors, (see 3.17 and 3.26).
q. When Government-furnished property is provided (see 3.17).

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6.3 First article. Any changes or deviations of production compressors from the approved first article model during production will be subject to the approval of the contracting officer. Approval of the first article model will not relieve the contractor of his obligation to furnish compressors conforming to this specification. The contractor should provide a written report to the contracting officer covering all deficiencies, malfunctions, and test failures. This report shall be presented within 48 hours from the time of failure and should describe components and parts affected, both directly and indirectly, corrective action taken or to be taken by the contractor, and the contractor's proposed method of proving the adequacy of the corrective action.

6.3.1 First article inspection. When a first article inspection is required, the items should be a preproduction compressor. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, tests and approval of the first article test results and disposition of the document's first article.

6.4 Lubricants. Selection of lubricants shall be in accordance with section 5 of MIL-STD-838. Unless otherwise authorized by the U.S. Army Belvoir Research and Development Center, lubricants for ground equipment systems shall be restricted to those listed in MIL-HDBK-113.

6.5 Government-furnished property. The contracting officer should arrange to furnish to the contractor the property listed in 3.17, 3.25.4 and 3.26.5 as required. The military design engine and information regarding the installation of the engine will be furnished by the Government (see 3.26.1).

6.6 Government-loaned property. The contracting officer should arrange for loan of the Government property specified in 3.18 as required.

6.7 Free air. Free air is defined as air at atmospheric conditions surrounding the compressor, at the point where the compressor is installed, but not in the immediate vicinity where it may be affected by the heat of the compressor or engine cooling system.

6.8 Information figures. Figures 1 and 2 show a handtruck-mounted compressor and a caster-mounted compressor, respectively, which have been found acceptable; however, the figures are included for illustration only and are not intended to preclude the furnishing of other designs which conform to this specification.

6.9 Cross-country terrain. Cross-country terrain shall consist of generally level or improved open fields, broken ground, loose sand, or gravel road (see 4.5.2.15).

6.10 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractor's maintenance and operation manual to be furnished with each compressor.

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6.11 Provisioning. The contracting officer should include provisioning requirements for repair parts and special tools as necessary, and instructions on shipment of compressors. A suggested paragraph is as follows:

"Shipment of compressors shall include repair parts, special tools, operational instructions, and accessories, unless exceptions are provided elsewhere in the contract."

6.12 Recovered material. It is encouraged that recovered material be used when practical as long as it meets the requirements of the specification (see 3.4).

6.13 Noise limits. If the compressor is designed to meet the noise limit requirement, the weight limit requirement may be exceeded. In view of this, the solicitation process must provide bidders an opportunity to:

a. Address what design aspects (design or shielding) will be required to meet the noise requirement and the cost of each design aspect. This information must be supplied with engineering feasibility data in accordance with MIL-STD-1474.
b. List the calculated weight increase for each noise suppression design aspect. These calculations must be provided for each design aspect.

Before an award is made, a decision on final noise and weight limits must be rendered in conjunction with the Belvoir R&D Center, ATTN: STRBE-NC, Fort Belvoir, Virginia 22060-5606.

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Navy - YD

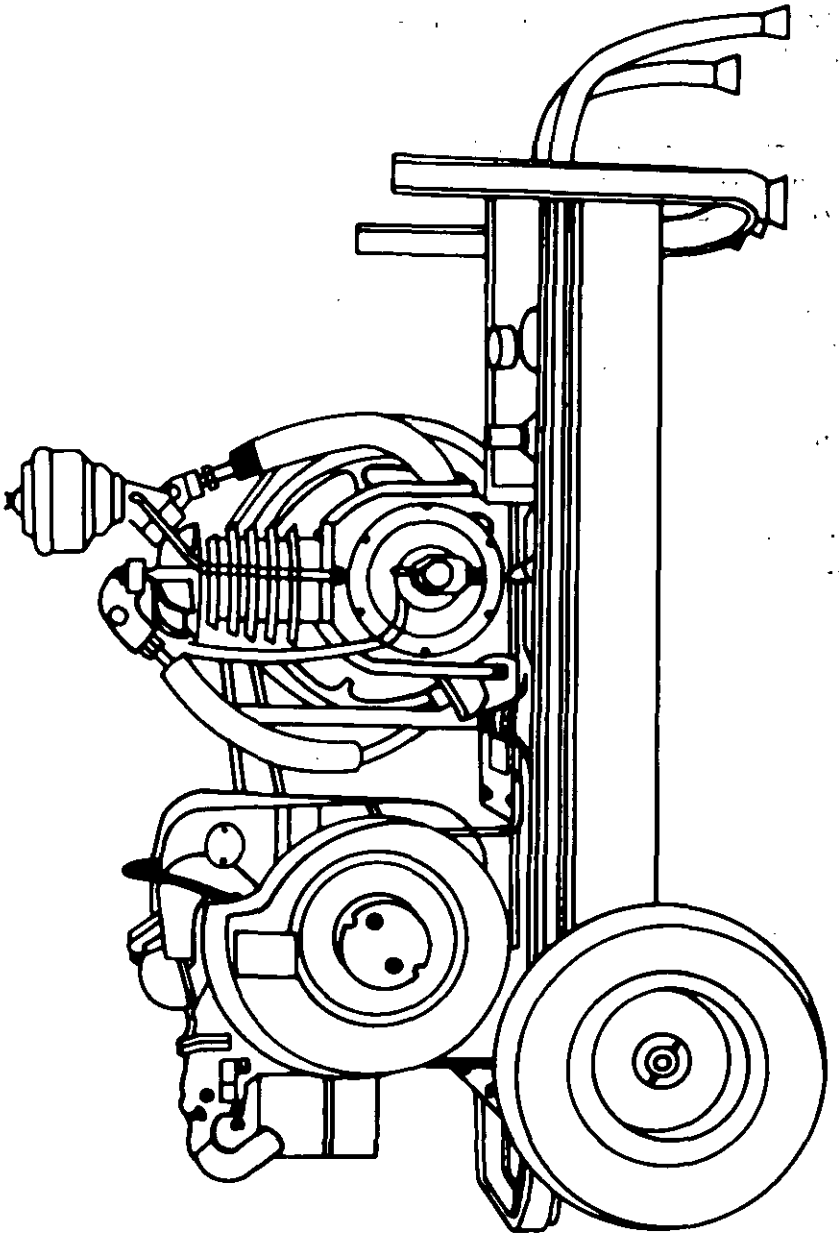
Review activity:
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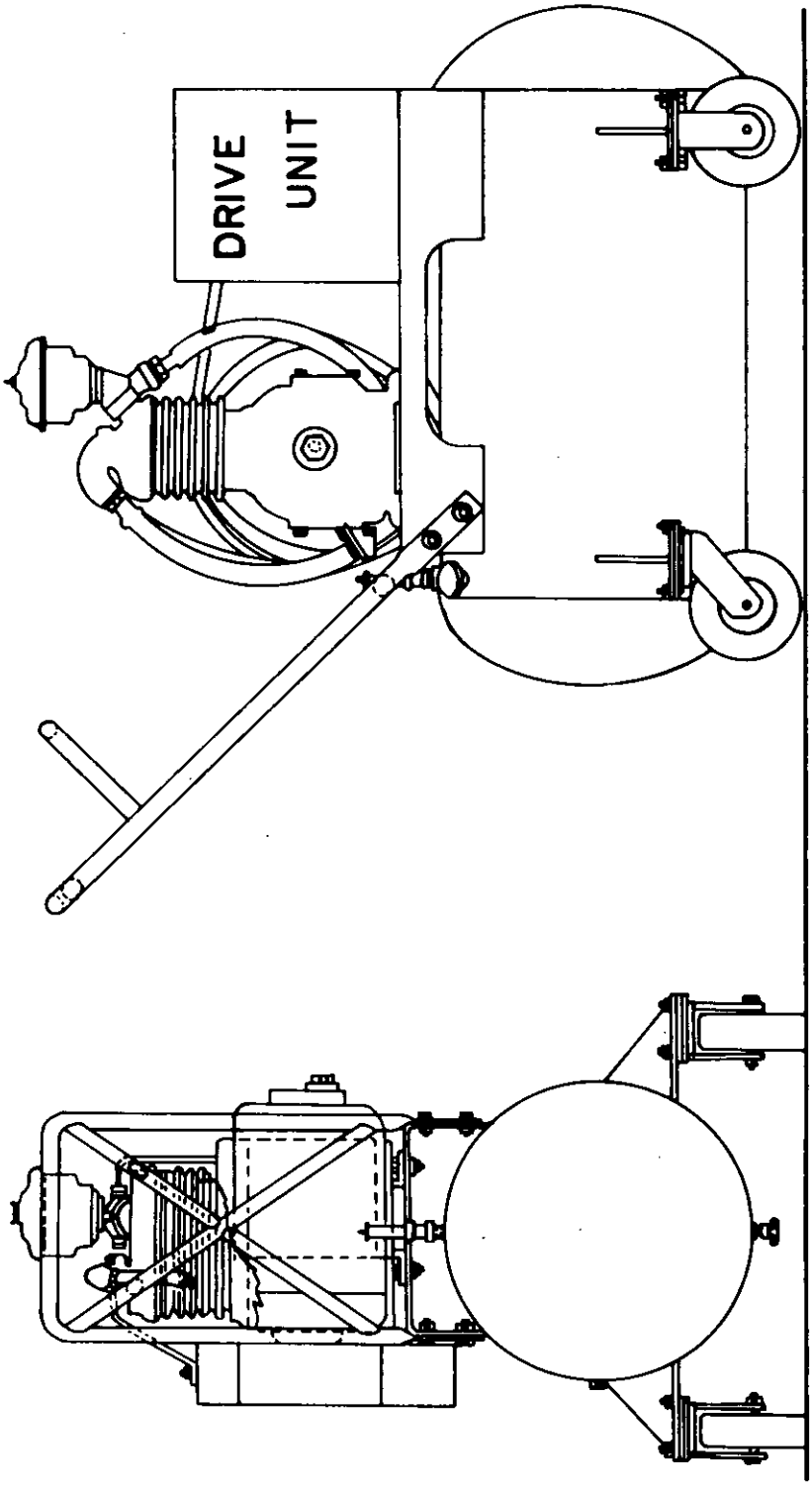
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X3640

FIGURE 1. Compressor, air, hand-truck-mounted (gen. configuration).

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FIGURE 2. Compressor, air, mounted (general configuration).

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DATA SHEET
MFG NAME_____ P/N_____ Record Time & Amount of 1. Engine Oil Added 2. Compressor Oil Added Sheet__of__

Test Title:_____ Run No._____ Date_____

	Item	SYM	UNITS	Readings 1	2	3	4	5	6	7	8	9	10	...	...		
1.	Time Start, Readings, Stop	R_t	HRS. MIN.	(IF STOP IS UNSCHEDULED GIVE REASON FOR STOP AND DOWN TIME)													
2.	Temp. (Ambient) Dry Bulb	T_a	°F														
3.	Temp. (Wet Bulb)	Twb	°F														
4.	Pressure (Ambient)	P_b	In. Hg														
5.	Temp., Compressor Lubricant	T_L	°F														
6.	Temp., Receiver Discharge	T_r	°F														
7.	Pressure, Rec.	T_r	PSIG														
8.	Pressure, Compressor Intake	P_a	PSIA														
9.	Temp., Compressor Intake	T_1	°F														
10.	Temp., Upstream of Nozzle	T_2	°F														
11.	Pres. Differential Across Nozzle	ΔP	$In.H_2O$														
12.	Comp. Speed	N	RPM														

FIGURE 3. Preproduction test log.

X-1904 A

MIL-C-52980A

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-52980A	Compressors, Reciprocating and Rotary Gasoline-Engine Driven,

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
Air	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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